DSC 291 — Report 1

Liver Disease Risk Prediction

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1. Model Selection

The updated model achieves a C-index score of .8766, surpassing the score of the baseline model: .8466. For our baseline, input features were preprocessed by imputing missing non-numerical values with a string value "missing" and One Hot Encoding all categorical variables. Numeric missing values were imputed with the column median. For our improved model, no changes were made to the pre-processing steps but the model chosen differed. Our baseline used a standard RandomForestRegressor, which did not account for the age at which patients had their respective event. It also handled censored patients (those who left the study before it was concluded or those whose data is unknown) as true negatives. For the improved model, we used a RandomSurvivalTree from sklearn.survival. This model incorporates the age at which each event occurred and handles censored patients correctly by only counting them as part of the risk set up until their last observed age. It uses the log-rank test to separate patients by their event-time distributions rather than by a scalar label, which directly aligns with the C-index metric used for evaluation.

2. C-index

Overall, the model archives a C-index of 0.7611 for hepatic events and 0.7404 for death, having a slightly better hepatic risk prediction.

The performance is similar for gender, although the hepatic C-index differs 0.7819 vs 0.7410, with death having similar results.

For age groups, the model performs better in predicting hepatic events for younger patients (0.779 vs 0.747), while death prediction is stronger in older patients (0.764 vs 0.616). This suggests that age influences the model differently depending on the outcome.

BMI also impacts performance: patients with lower BMI show significantly better hepatic prediction (0.827 vs 0.701), while death prediction remains relatively stable.

Stratification by T2DM revealed more pronounced differences. For hepatic events, performance was relatively consistent between patients with and without T2DM (C-index $\approx$ 0.68–0.69). However, for death prediction, the model performed notably worse in patients with T2DM (C-index = 0.647) compared to those without (0.722). This indicates that the model may have reduced ability to capture mortality risk in diabetic patients.

3. Additional analysis

Calibration Analysis

We also performed a decile calibration analysis to evaluate the model's clinical utility. We separated the patients into ten equal sections (deciles) based on their predicted mortality risk. It ranged from the lowest 10% (Decile 0) to the highest 10% (Decile 9) and compared these predictions against the actual outcomes. We observed 2 phenomena:

The model successfully identifies a large safety zone, where patients in deciles 0 through 6 had a 0.00% actual death rate. This demonstrates that the model is highly reliable at identifying healthy individuals.

There is a high risk concentration in decile 9, with the actual death rate reaching 55.93%. This means that the model has successfully identified a specific, high risk group where the majority of actual clinical events are concentrated.

To conclude, this test has shown us that the doctors using this tool, could safely de-prioritize monitoring patients in deciles 0–6, and focus their resources on decile 9.

Comparison of Model Architectures

Our team iterated through three distinct architectures to capture the relationships between the features and decide on what model works best.

Subgroup	Value	N Patients	N Hepatic	N Death	Hep C-Index	Death C-Index
overall	all	1253	47	76	0.7611	0.7404
gender	1	655	25	44	0.7819	0.7375
gender	2	598	22	32	0.7410	0.7434
T2DM	0	522	18	28	0.6797	0.7223
T2DM	1	368	26	42	0.6940	0.6473
T2DM	2	97	2	6	0.9214	0.7950
Hypertension	0	460	16	24	0.7451	0.6689
Hypertension	1	426	28	46	0.6705	0.7113
Hypertension	2	99	2	6	0.8871	0.7643
Dyslipidaemia	0	491	23	31	0.6829	0.6629
Dyslipidaemia	1	378	20	37	0.7440	0.7292
Dyslipidaemia	2	117	3	8	0.8695	0.7997
bariatric_surgery	0	827	40	67	0.7500	0.7218
bariatric_surgery	1	72	4	3	0.4604	0.5238
bariatric_surgery	2	78	1	5	0.8571	0.6773
AgeGroup	old	596	27	53	0.7466	0.7637
AgeGroup	young	657	20	23	0.7791	0.6164
BMIGroup	high	427	16	32	0.7013	0.7100
BMIGroup	lowI	427	13	24	0.8270	0.7320

Table 1: C-index results stratified by subgroup.

Random forest

Score: 0.8466

We used it to establish a robust baseline for handling tabular data with missing values.

XGBoost

Score: 0.8620

Utilized gradient boosting to improve accuracy by focusing on hard to predict errors. It performed better than the random forest but not as good as Random Survival Forest.

Random survival forest

Score: 0.8764

This was the final model chosen. We chose it because of its ability to handle missing data. By using the estimators, it correctly accounts for patients who haven't experienced an event yet, rather than treating them as simple binary outcomes.

4. Future Steps

With more time, there are a few approaches we can take to improve our model. The first is more in-depth preprocessing. Our current approach is minimal, imputing with "missing" or column median. We could go further by using iterative or KNNimputation, removing columns with low variance, or a missingness column. Usually, a missing value implies a test wasn't ordered, which is meaningful.

We could also perform some level of hyperparameter tuning. Using GridSearch Cross Validation, we could likely find a set of hyperparameters that is optimal for the problem at hand.